

*MA.7.GR.2.3***Benchmark**

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

Benchmark Clarifications:

Clarification 1: Within this benchmark, the expectation is not to memorize the volume formula for a right circular cylinder or to find radius as a missing dimension.

Clarification 2: Solutions may be represented in terms of pi (π) or approximately.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.1
- MA.7.AR.2.2
- MA.7.AR.4.5
- MA.7.GR.1.4
- MA.7.GR.2.1, MA.7.GR.2.2

Terms from the K-12 Glossary

- Cylinder (Circular)
- Pi (π)
- Volume

Vertical Alignment**Previous Benchmarks**

- MA.6.GR.2.3
- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3

Next Benchmarks

- MA.912.GR.4.5

Purpose and Instructional Strategies

In previous courses, students solved problems involving volumes with a focus of right rectangular prisms. In grade 7 accelerated, students solve problems involving the volumes of right circular cylinders. In Geometry, students will solve problems involving the volume of cylinders, pyramids, prisms, cones and spheres.

- Instruction builds on students' knowledge of finding the volume of a rectangular prism, which also includes the area of the base multiplied by the height.
- Instruction encourages students to make conjectures about how to calculate the volume of a right circular cylinder. Students are provided the radius and height of several cylinders to verify or revise their conjecture (*MTR.6.1*).
- Instruction includes physical or virtual representations for the volume of a right circular cylinder (*MTR.2.1*).
 - For example, stack quarters one at a time to show repeated addition (the height) on the area of the base (the first quarter used).
- Instruction focuses on real-world situations to reinforce conceptual understanding of volume versus surface area (*MTR.7.1*).

Common Misconceptions or Errors

- Students often confuse the vocabulary base, length, height and “B” (base area), when moving between two- and three-dimensional figures. To address this misconception, continue to use the parts of the net to calculate the surface area, rather than focusing on the formula.
- Students may incorrectly believe that whatever is lying flat is the base of the figure. To address this misconception, remind students that while a cylinder may lay on its side, the bases are the parallel circles with the height being the perpendicular distance between them. Provide multiple orientations of objects and continue to break them down to their nets.
- Students may incorrectly apply the formulas for area, surface area and volume. To address this misconception, use these concepts in context, or a manner in which they understand the meaning behind the terms to foster their conceptual development (MTR.7.1).
Students may incorrectly invert the height and base of a cylinder when the cylinder is presented horizontally.

Strategies to Support Tiered Instruction

- Instruction includes the use of geometric software to allow students to explore the difference between base, length, height and “B” (base area).
- Teacher creates and posts an anchor chart with visual representations of a right circular cylinder to assist in correct academic vocabulary when solving real-world problems.
- Teacher provides students with an example of a three-dimensional figure in its original position then provides multiple orientations to discuss how the location of the figure’s base changes, but the dimensions of the figure do not change.
 - For example, two right circular cylinders are shown below with the same dimensions but in different orientations. The base is highlighted in each.



- Teacher co-creates a visual of a three-dimensional figure and its dimensions in the context of a real-world problem, and color-coding and labeling the dimensions.
- Instruction includes opportunities for students to solve for the volume of a given right circular cylinder in terms of pi before replacing the value of pi with an approximation to determine the estimated volume.
- Teacher provides instruction focused on manipulatives or geometric software for students to develop understanding of the difference between the formulas for area, surface area and volume.

- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Teacher encourages students to continue to use the parts of the net to calculate the surface area, rather than focusing on the formula.
- Provide multiple orientations of cylinders, focusing on the height as the perpendicular distance between the parallel bases, continuing to break them down to their nets.

Instructional Tasks

Instructional Task 1 (MTR.1.1)

Coffee2Go wants to build a record-breaking giant coffee cup for a promotional celebration at a convention for coffee drinkers. In 2020, the Guinness Book of World Records reported that the World's Largest Cup of Coffee contained 2,010 gallons of fresh-brewed coffee brewed.

Part A. What questions would need to be answered to approach this problem? Do you have all the information you need to solve the problem? Why or why not?

Part B. The Guinness Book of World Records, in 2020, documented the Largest Coffee Cup with a height of 8 feet and a diameter of 8 feet. How many cubic feet of coffee can be held in the World's Largest Coffee Cup?

Part C. There are approximately 1.25 gallons of coffee in a cubic foot. If Coffee2Go has a goal of creating a coffee cup that holds 2,410 gallons of coffee, what could be the height and diameter of the coffee cup if they are the same size?

Instructional Task 2 (MTR.7.1)

Karim is setting up an inflatable pool in the yard for the little kids to play. The pool measures $3\frac{1}{2}$ feet across and 16 inches deep. What is the volume of water needed to fill the pool 2 inches from the top? Justify your solution.

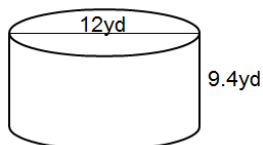
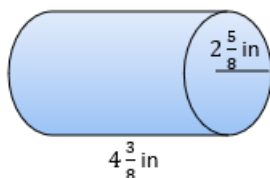
Instructional Items

Instructional Item 1

What is the height of a cylinder, with a radius of 3.4 meters (m), whose volume is 198 m^3 ? Round to the nearest hundredth.

Instructional Item 2

Determine the exact volume of each of the following cylinders.

*Instructional Item 3*

A candle mold has 24π in³ of space in which wax can be poured to make a candle. If the radius of the mold is 2 inches (in), what is its height?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*